Evolution of Sentiment Analysis: From Lexicon-Based and Traditional Machine Learning Methods to Transformers and Large Language Models

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***Abstract*—**Sentiment analysis has evolved from lexicon-based rules and sparse feature representations to deep sequential networks, Transformer encoders, and large language models (LLMs). This mini-survey reviews five recent studies published between 2022 and 2024 to examine that evolution through a common application: sentiment classification. The reviewed work spans TextBlob-assisted annotation, bag-of-words and TF-IDF features, trigram and fastText representations, LSTM/GRU/BiLSTM networks, BERT-based contextual models, and modern LLMs evaluated in zero-shot, few-shot, and cross-lingual settings. The comparison shows three broad shifts: representation moved toward learned contextual representations; supervision moved from task-specific training toward pretraining, fine-tuning, and in-context learning; and evaluation expanded from single-domain classification to multi-task and multilingual settings. A closer reading also shows that headline gains can be inflated by automated labels and weak baselines. The most credible BERT advantage in the reviewed conventional studies is a modest 1–5 percentage points, while the LLM studies show that adaptation method can matter as much as model size. Annotation quality, prompt design, language coverage, computational cost, and task complexity remain important limitations. Overall, the LLM era changes how sentiment models are adapted and used, but it extends rather than eliminates earlier approaches.

***Index Terms*—**sentiment analysis, TextBlob, n-gram, LSTM, BERT, Transformer, large language model, zero-shot learning, few-shot learning, cross-lingual NLP.

I. INTRODUCTION

Sentiment analysis is a natural language processing (NLP) task that identifies attitudes, opinions, or affective polarity expressed in text. In practical applications, it is commonly used to classify text as positive, negative, or neutral, although modern sentiment analysis also includes aspect-level opinion extraction, emotion recognition, stance detection, hate-speech detection, irony detection, and other forms of subjective-language analysis. The problem is important because large volumes of user-generated text are produced through social media, product reviews, customer-service interactions, and online forums. Manual analysis of such text does not scale well, which has motivated the use of computational methods for automatic sentiment classification [1], [3], [4].

The technical history of sentiment analysis mirrors the broader evolution of NLP. Earlier approaches depended heavily on lexicons and manually designed features. A lexicon-based system assigns sentiment scores to words or phrases and combines those scores to estimate the polarity of a sentence or document. TextBlob is one example of this family. Traditional supervised machine-learning systems then represented documents through bag-of-words (BoW), term frequency-inverse document frequency (TF-IDF), or n-gram features and trained classifiers such as support vector machines, logistic regression, decision trees, or ensemble methods [1], [2]. These techniques can be effective, but their representation of language is limited because a word or n-gram is usually treated as a fixed feature and the broader sentence context is only indirectly represented.

Deep learning changed this design by learning text representations and classification patterns jointly. Recurrent neural networks (RNNs), long short-term memory (LSTM) networks, gated recurrent units (GRUs), and bidirectional LSTMs (BiLSTMs) improved the ability to model ordered sequences and longer dependencies. Nevertheless, recurrent architectures process tokens sequentially and can still have difficulty representing complex context efficiently. Transformer-based models such as BERT introduced self-attention and contextual token representations, allowing the meaning of a token to depend on the surrounding sentence. Across the reviewed studies, BERT-based representations show consistent within-study improvements over static or local representations, although the magnitude of the improvement differs substantially across experimental designs [2], [3].

The emergence of LLMs extends this progression further. Instead of requiring a new classifier to be trained from scratch for every task, an LLM can often perform sentiment analysis through a natural-language instruction, sometimes with no labeled examples (zero-shot) or a small number of demonstrations (few-shot). This changes both the technical workflow and the research question. The issue is no longer only which supervised classifier produces the highest accuracy; it is also when prompting, in-context learning, fine-tuning, smaller specialized models, or multilingual models are the most appropriate choices [4], [5].

This report asks: How has sentiment analysis evolved from lexicon-based and traditional machine-learning methods to Transformer-based models and modern LLMs, and what

performance and methodological trade-offs have emerged across these approaches? Five recent papers are reviewed comparatively. The analysis uses six common dimensions: (1) research problem and dataset, (2) representation or model family, (3) training or adaptation method, (4) evaluation metrics, (5) key reported result, and (6) methodological strengths and limitations.

II. BACKGROUND: EVOLUTION OF SENTIMENT ANALYSIS

A. Lexicon-Based and Sparse Feature Methods

Lexicon-based sentiment analysis estimates polarity from dictionaries in which words are associated with sentiment values. Aljedaani et al. [1] use TextBlob as a central example and compare it with VADER and AFINN. Such methods have practical advantages: they are simple to use, comparatively transparent, inexpensive to run, and do not require the same training process as supervised neural models. However, they are sensitive to vocabulary coverage, negation, informal language, ambiguity, domain-specific meaning, and context. A word may carry different sentiment in different contexts, and a simple polarity score may miss that distinction.

Traditional machine-learning systems reduce text to numerical features before classification. BoW counts token occurrences, TF-IDF reweights terms according to their relative importance, and n-gram methods represent short sequences of adjacent tokens or characters. These representations can then be used with classifiers such as support vector machines (SVMs), logistic regression, decision trees, random forests, gradient boosting, or extra-trees ensembles. Paper [1] evaluates BoW and TF-IDF with multiple supervised classifiers, while paper [2] uses trigram features with fastText and a linear SVM. These methods remain useful baselines because they are computationally efficient and often competitive for well-defined classification problems. Their central limitation is that the representation is relatively local or static: a feature records the presence or frequency of text patterns without fully modeling the changing meaning of words across a sentence.

B. Recurrent Deep Learning

RNN-based models introduced a learned sequential representation. Instead of treating a document as an unordered vector of features, an RNN processes tokens in sequence and carries information forward through a hidden state. LSTM and GRU architectures were designed to preserve useful information across longer spans and reduce the difficulty of learning long-term dependencies. BiLSTM extends this idea by processing the sequence in both directions. In the reviewed studies, LSTM, GRU, CNN-LSTM, LSTM-GRU, and fastText-BiLSTM systems represent the transition from feature-engineered machine learning to deep learned representations [1], [2]. Their performance demonstrates that learned sequence representations can outperform traditional feature pipelines when sufficient training data are available.

The benefits of recurrent models do not remove all limitations. Sequential processing restricts parallelism, and static embeddings such as Word2Vec still assign the same underlying vector to a word regardless of sentence context. Bello et al. [3] emphasize this weakness when contrasting Word2Vec-based models with BERT. The difference is important for sentiment because polarity can depend on contrast, negation, sarcasm, targets, or contextual shifts that cannot be captured by a single fixed word vector.

C. Transformers and BERT

Transformer models replace recurrence with attention mechanisms that allow the model to relate tokens directly across a sequence. BERT uses a bidirectional Transformer encoder and is pre-trained on large text corpora before being fine-tuned for a downstream task. This architecture gives each token a contextual representation that depends on both preceding and following text. Papers [2] and [3] use BERT as the key transition point from recurrent or static-embedding systems to pretrained contextual language models. In both studies, BERT-based systems outperform the corresponding trigram, fastText, or Word2Vec baselines within the reported experiments.

D. Large Language Models and In-Context Learning

LLMs build on the Transformer paradigm but change how a model is adapted to a task. In-context learning specifies task behavior at inference time by placing instructions and, when needed, labeled demonstrations in the prompt rather than updating model parameters. The labels “zero-shot” and “few-shot” are not used identically in the two LLM studies reviewed here. In Zhang et al. [4], zero-shot means an instruction with no demonstrations, while few-shot means that demonstrations are added to the prompt. In Zhu et al. [5], “zero-shot cross-lingual” means that no examples from the target language are provided: public models are still fine-tuned on 30,000 English examples, and proprietary GPT models are prompted with 300 English source-language examples. Their “few-shot” setting adds 60, 150, or 600 target-language examples, either through supervised fine-tuning for public models or by appending examples to the GPT prompt. These definitions must be kept separate when interpreting results. Fig. 1 summarizes both the model progression and the dominant adaptation method at each stage.

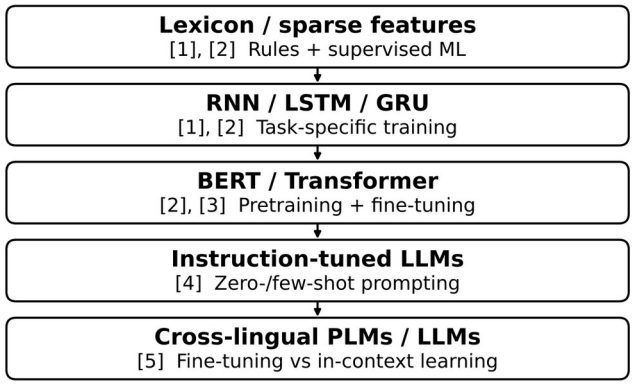


Fig. 1. Evolution from feature-based sentiment analysis to contextual and instruction-adapted models; paper numbers indicate where each stage is examined.

E. Why N-gram and Feature-Based Methods Still Matter

Although the main direction of this report moves toward Transformers and LLMs, n-gram and sparse-feature methods should not be treated as obsolete. They provide an

interpretable reference point for understanding what later representations add. A unigram represents individual tokens, a bigram captures adjacent pairs, and a trigram captures three-token sequences. Such local features can encode cues such as "not good" or "very helpful" without requiring a large pretrained model. Sparse and n-gram systems are also comparatively easy to inspect, reproduce, and deploy on limited hardware, which makes them useful baselines for deciding whether additional model complexity is justified.

Feature-based models also have practical benefits for debugging. A practitioner can inspect the vocabulary, feature weights, and misclassified examples more directly than in a very large neural model. In regulated, low-resource, or latency-sensitive settings, this transparency can be valuable. Moreover, sparse models can serve as sanity checks. If a very large model only marginally outperforms a simple TF-IDF/SVM baseline, the added complexity may not be justified. The progression toward LLMs does not imply that earlier methods disappear; instead, they become baselines and specialized tools within a larger model-selection toolkit.

F. Evaluation Metrics and What They Measure

Accuracy, precision, recall, and F1-score answer different evaluation questions. Accuracy is the proportion of all predictions that are correct, but it can be misleading under class imbalance. Precision measures how often a predicted class is correct, recall measures how much of the true class is recovered, and F1 combines precision and recall. Macro-F1 gives each class equal weight, whereas micro-F1 aggregates decisions across instances. Structured sentiment tasks may require exact or linked predictions rather than a single class label, so task-specific metrics are necessary when the output structure changes.

Metric choice is part of the task definition. A model can have strong general language ability yet score poorly when a structured ABSA prediction is counted as correct only if all required elements match the reference. Conversely, high binary classification accuracy does not demonstrate fine-grained aspect understanding. The evaluation measure must match the behavior the system is expected to produce.

G. Training, Fine-Tuning, and Prompting as Different Adaptation Paradigms

Traditional supervised sentiment models learn task parameters directly from labeled examples. In an SVM or logistic-regression pipeline, the learned decision boundary maps feature vectors to sentiment classes. In an LSTM or GRU, the network parameters and embeddings are optimized through backpropagation for the target labels. BERT changes this process by beginning with a pretrained language representation and then fine-tuning the pretrained parameters and classification layer on the target dataset. Because the model starts with broad linguistic knowledge, it can learn the downstream decision function from fewer task-specific examples than a model trained from random initialization.

Prompting is an adaptation mechanism rather than a model architecture. It can reduce the need to retrain model parameters, but performance may depend on wording, demonstration selection, output formatting, and context-window capacity. Fine-tuning instead changes model parameters using labeled examples. Because model family and adaptation method can vary together, an observed score difference may reflect both the underlying representation and the way the model was adapted.

III. REVIEW SCOPE AND COMPARATIVE FRAMEWORK

Candidate studies were identified through publisher and academic databases including ScienceDirect, IEEE Xplore, MDPI, and the ACL Anthology. Search combinations included "sentiment analysis TextBlob deep learning," "BERT sentiment analysis," "large language models sentiment analysis," and "cross-lingual sentiment analysis." Inclusion criteria were: publication from 2022–2024, peer-reviewed empirical work, an explicit sentiment-analysis task, reported datasets/models/evaluation metrics, and coverage of a distinct stage in the transition from lexicon or feature-based methods to contextual Transformers and LLMs. To keep the comparison focused, five studies meeting these criteria were retained.

Each paper is compared using the same six dimensions introduced in Section I: research problem and dataset; representation or model family; training or adaptation method; evaluation metrics; key reported result; and methodological strengths or limitations. Table I summarizes these dimensions. The later comparative analysis uses the same framework rather than constructing a cross-paper leaderboard, because the datasets, label schemes, and evaluation protocols differ.

IV. OVERVIEW OF SELECTED PAPERS

A. Aljedaani et al. (2022): TextBlob, Machine Learning, and Deep Learning

Aljedaani et al. [1] investigate sentiment analysis for airline-related tweets and focus on a problem that is sometimes ignored in model comparisons: the quality of the labels used for training. Their principal dataset is the Kaggle "twitter-airline-sentiment" dataset containing 14,640 tweets about six U.S. airlines with positive, negative, or neutral labels. The authors observed apparent inconsistencies in some original labels and tested whether relabeling with TextBlob changes downstream model performance, while also comparing TextBlob with VADER and AFINN.

The study combines several generations of sentiment technology in one framework. For traditional machine learning, BoW and TF-IDF features are used with random forest, extra trees, gradient boosting, decision tree, logistic regression, and support vector classifiers. For deep learning, the authors evaluate CNN, LSTM, GRU, CNN-LSTM, and a proposed stacked LSTM-GRU model. The LSTM-GRU system contains an embedding layer followed by LSTM and GRU components and a softmax classifier for the three sentiment classes. Accuracy, precision, recall, and F1-score are used as the principal performance measures.

The highest reported 0.97 accuracy for LSTM and LSTM-GRU is measured against TextBlob-assigned labels, not against human-expert ground truth. LSTM-GRU also reports macro precision and F1 of 0.96 under that labeling scheme. This distinction is critical because a separate 1,000-tweet experiment found agreement between TextBlob and human

experts on only 491 tweets, with 509 disagreements. The result is best interpreted as high agreement with a rule-based labeling function rather than 97% human-validated sentiment accuracy. The authors themselves state that TextBlob should assist rather than replace human annotation [1].

This paper is important because it shows that model performance can be dominated by the definition and quality of the target labels. A classifier can obtain a high score by reproducing a consistent automated labeling rule even when that rule disagrees substantially with human judgment. The study contributes more strongly as evidence about annotation design than as evidence that the LSTM-GRU architecture achieves near-human sentiment understanding.

B. Chinnalagu and Durairaj (2022): Trigram and BiLSTM to Fine-Tuned BERT

Chinnalagu and Durairaj [2] present a controlled comparison among four sentiment classifiers: a fine-tuned BERT-base-cased model, a hybrid fastText-BiLSTM model, a fastText trigram model, and a linear SVM with trigram features. Their experiments use Amazon and Yelp customer-review datasets collected from Kaggle. In total, 778,631 reviews are used, with 545,042 records (70%) allocated to training and 233,589 (30%) to evaluation. Text is cleaned by lowercasing and removing HTML tags, repeated headings, special characters, and stop words. The output is binary positive/negative sentiment.

This study provides a useful bridge across three representation styles. LSVM-Trigram is a traditional sparse/local feature model. fastText-Trigram uses subword n-gram embeddings, and the hybrid fastText-BiLSTM model adds bidirectional sequential modeling. Fine-tuned BERT introduces contextual Transformer representations. The authors evaluate accuracy, recall, precision, and F1. Fine-tuned BERT achieves 91.0% accuracy with recall 0.91, precision 0.90, and F1 0.90. Hybrid fastText-BiLSTM achieves 90.0% accuracy and F1 0.89, fastText-Trigram reaches 88.0% accuracy and F1 0.88, and LSVM-Trigram records 86.11% accuracy and F1 0.85 [2].

The result supports the value of contextual pretraining, but the study also highlights a resource trade-off. BERT training and evaluation are performed in Google Colab, whereas fastText, BiLSTM, and LSVM are run on a standard Windows server with an Intel i7 processor and 16 GB of memory. The authors note that the hybrid fastText-BiLSTM model has lower computational cost than BERT while remaining close in predictive performance. This is an important counterpoint to a simple "newer is always better" narrative: a one-point accuracy improvement may or may not justify increased computational requirements depending on the deployment context.

C. Bello et al. (2023): Contextual BERT Versus Word2Vec-Based Deep Models

Bello, Ng, and Leung [3] investigate sentiment classification of tweets using BERT combined with CNN, RNN, and BiLSTM classifiers. Six Kaggle Twitter datasets are concatenated; the paper reports about 213k records, although counts vary slightly between its text (212,661 rows) and Table 1 (213,355 rows). The final class distribution is approximately 40.1% positive, 33.7% neutral, and 26.2% negative.

The experiment compares Word2Vec- and BERT-based representations paired with CNN, RNN, and BiLSTM classifiers. Reported accuracy is 57% for Word2Vec-CNN, 48% for Word2Vec-RNN, and 55% for Word2Vec-BiLSTM, while all three BERT variants report 93% accuracy [3]. The raw gap is large, but the Word2Vec baseline is methodologically weak: the majority class alone accounts for 40.1% of the data, so the 48% Word2Vec-RNN result is only modestly above a trivial majority prediction. The Word2Vec deep models are trained for 10 epochs with a learning rate of 1×10^{-5}, an unusually conservative setting for models trained from scratch, and the paper does not clearly state the train/test split or report repeated runs, seeds, or variance. These choices make it plausible that the baseline is undertrained and that the apparent 36–45 point BERT advantage is inflated.

The contextual explanation remains plausible: BERT forms token representations from surrounding text, whereas a static Word2Vec embedding assigns one vector per word. However, the evidence from [3] should support the direction of the advantage more strongly than its magnitude. Paper [2] provides a more conservative comparison on a clearly stated 70/30 split, where fine-tuned BERT reaches 91% accuracy versus 90% for fastText-BiLSTM, 88% for fastText-Trigram, and 86.11% for LSVM-Trigram. Thus, a roughly 1–5 percentage-point advantage in [2] is a more credible estimate of the practical BERT gain under that study's setup. Bello et al. [3] suggest as future work additional data sources, richer emotion categories, and other Transformer models such as RoBERTa; the paper does not present a dedicated limitations section.

D. Zhang et al. (2024): Sentiment Analysis in the LLM Era

Zhang et al. [4] move beyond a single sentiment-classification dataset and conduct a broad "reality check" of LLM capabilities. They evaluate 13 tasks across 26 datasets and group the tasks into three categories: sentiment classification (SC), aspect-based sentiment analysis (ABSA), and multifaceted analysis of subjective text (MAST). SC includes document-, sentence-, and aspect-level classification using datasets such as IMDb, Yelp, MR, SST, Twitter, Laptop14, and Restaurant14. ABSA includes unified aspect-based sentiment analysis, aspect-sentiment triplet extraction, and aspect sentiment quadruple prediction. MAST covers tasks such as implicit sentiment, hate speech, irony, offensive language, stance, comparative opinion, and emotion recognition. A maximum of 500 examples is sampled from each test set for balanced evaluation across tasks.

The LLMs include Flan-T5, Flan-UL2, text-davinci-003, and ChatGPT, while T5-large is a smaller model trained on in-domain labeled data. Under zero-shot prompting, ChatGPT averages 78.24 on sentiment classification versus 80.65 for fully trained T5-large, but the gap widens on structured ABSA (34.47 versus 61.06). Flan-UL2 illustrates the structured-output problem even more sharply: it averages 77.31 on sentiment classification and 68.33 on MAST, yet scores 0.00 on the ABSA group in the reported zero-shot evaluation [4].

The evidence supports a distinction between broad classification ability and reliable generation of task-specific structured outputs.

The few-shot results change the comparison. With only a small number of examples, LLMs generally outperform T5 trained on the same limited amount of data. For example, in the one-shot setting ChatGPT scores 81.47 on document-level sentiment classification compared with 66.76 for T5, and the advantage is much larger on the structured UABSA, ASTE, and ASQP tasks. The authors also find that the effect of additional demonstrations is task dependent: more examples help complex ABSA tasks, but they can offer limited or even negative benefit on some MAST tasks. Prompt wording and output format matter, especially for complex tasks [4].

A second contribution of the paper is SENTIEVAL, a benchmark designed to evaluate sentiment ability more comprehensively through varied natural-language task instructions. The benchmark responds to a limitation of current evaluation practice: studies often use different tasks, datasets, and prompts, which makes it difficult to judge whether a model has broad sentiment competence. The paper itself also acknowledges limitations. The 13 tasks and 26 datasets are not exhaustive, and the evaluated datasets are English-only, leaving language- and culture-specific sentiment behavior unresolved [4].

E. Zhu et al. (2024): Cross-Lingual Sentiment Analysis with Small and Large Models

Zhu et al. [5] investigate cross-lingual sentiment analysis using proprietary call-center transcripts in English, Spanish, French, and Chinese. English provides 30,000 fine-tuning examples and 3,000 development examples; each target language provides 600 examples for fine-tuning and 3,000 for testing. Spanish is sampled and annotated directly. French and Chinese are machine-translated from the annotated English data using GPT-4 and then reviewed by target-language speakers. Because GPT-4 is later evaluated on those translated test sets, this creates a methodological dependency between data generation and evaluation that should be kept in mind, although human review mitigates obvious translation errors.

The model comparison includes multilingual SMLMs (mBERT, XLM-R, mT5), public English-centric LLMs (Mistral, Falcon, Llama-2, Llama-3), and GPT-3.5/GPT-4. The paper's "zero-shot cross-lingual" setting does not mean no examples overall: public models are fine-tuned on the 30,000 English examples, while proprietary GPT models receive 300 balanced English examples in context; zero-shot refers specifically to having no target-language examples. Under that setup, GPT-3.5 and GPT-4 obtain average F1 scores of 79.4 and 78.1, respectively, while mT5-large reaches 73.5 and is the strongest public model [5].

The few-shot experiment adds the same number N of target-language examples, with N = 60, 150, or 600. Public models use those examples for supervised fine-tuning together with the English data, whereas GPT-3.5/4 append them to the prompt. This distinction produces one of the paper's clearest findings: GPT-4 changes from 79.2 average F1 at 60-shot to 78.6 at 150-shot and 78.0 at 600-shot, while several fine-tuned public models rise into roughly the 84–85 F1 range at 600 shots. The target-language examples materially improve parameter-updated public models but provide little benefit to GPT-4 when supplied only as additional in-context demonstrations. The result strengthens the argument that adaptation method can matter as much as model size [5].

TABLE I. Comparative summary using the six-dimension review framework.

Study	Problem / Data	Model / Representation	Adaptation	Metrics & Key Result	Critical Limitation
[1]	14,640 U.S.-airline tweets; 3-class SA	TextBlob/VADER/AFINN; BoW/TF-IDF+ML; LSTM/GRU	Supervised on original or TextBlob labels	Accuracy, P/R/F1; 0.97 accuracy and 0.96 F1 against TextBlob labels	0.97/0.96 are agreement with TextBlob labels; automated labels diverge from human judgment
[2]	778,631 Amazon/Yelp reviews; binary SA	LSVM/fastText trigram; fastText-BiLSTM; BERT	Supervised training / BERT fine-tuning	Accuracy, P/R/F1; BERT 91%, BiLSTM 90%, fastText 88%, LSVM 86.11%	BERT gain is modest (1–5 points) and costs more compute than the BiLSTM hybrid
[3]	Six Kaggle Twitter datasets; ~213k; 3 classes	Word2Vec or BERT + CNN/RNN/BiLSTM	Task training / BERT fine-tuning	Accuracy, P/R/F1; BERT 93%, Word2Vec 48–57%	Weak Word2Vec baseline; split and variance reporting are unclear
[4]	13 tasks, 26 English datasets: SC, ABSA, MAST	Flan-T5/UL2; GPT-3.5 family; T5-large	Zero-/few-shot prompting vs supervised T5	Accuracy/Micro-F1; strong SC/low-shot results, but specialized T5 stronger on structured ABSA	Prompt/output sensitivity; English-only evaluation; task set is not exhaustive
[5]	Call-center transcripts: EN→ES/FR/ZH	mBERT/XLM-R/mT5; Mistral/Falcon/Llama; GPT-3.5/4	Public models fine-tuned; GPT in-context	Micro-F1; GPT leads with no target-language examples; fine-tuned public models lead as target data is added	"Zero-shot" still uses English supervision; FR/ZH are GPT-4 translations

V. COMPARATIVE ANALYSIS

A. Evolution of Data and Task Scope

The five studies show a clear expansion in the scope of sentiment analysis. Paper [1] focuses on one domain, one platform, one language, and three sentiment classes. Paper [2] expands the training volume through two customer-review sources but still treats sentiment as binary classification. Paper [3] increases diversity by combining six Twitter datasets while retaining the positive/neutral/negative formulation. Papers [4] and [5] move beyond the traditional single-dataset classifier: [4] evaluates 13 tasks and 26 datasets, and [5] tests cross-lingual transfer across three target languages. This progression matters because a model that performs well on one balanced or domain-specific classification dataset may not generalize to

structured sentiment extraction, conversational data, cultural variation, or low-resource languages.

The data also become more difficult in ways that accuracy alone cannot capture. Twitter data contain informal language, abbreviations, hashtags, and ambiguous expressions. Customer reviews are longer and domain-dependent. Structured ABSA requires the model to identify multiple linked sentiment elements rather than only a class label. Call-center transcripts contain disfluencies and automatic speech-recognition errors. The move to broader tasks changes the meaning of "good performance." A 93% accuracy result on a three-class Twitter classification problem cannot be compared directly with a Micro-F1 score on aspect triplet extraction or a cross-lingual F1 score on spoken-conversation transcripts.

B. Evolution of Representation

The most visible technical shift is from explicit feature construction to learned contextual representation. In paper [1], BoW and TF-IDF summarize a document using word occurrence statistics. TextBlob adds a lexicon-based polarity rule. In paper [2], trigram features and fastText introduce local subword information, while BiLSTM learns sequential dependencies. In paper [3], Word2Vec provides static semantic vectors, but BERT replaces them with contextual vectors derived from a bidirectional Transformer. Papers [4] and [5] then use general pretrained language models in which the same model can respond to a variety of task instructions without a dedicated feature engineering pipeline.

This evolution can be interpreted as a gradual reduction in the amount of task-specific representation engineering required from the practitioner. BoW requires explicit token counting; TF-IDF requires feature weighting; n-gram methods require choosing a local context width; Word2Vec learns embeddings but still fixes one vector per word; LSTM/BiLSTM learns sequence behavior but still depends on task training; BERT provides pretrained contextual representations that can be fine-tuned; LLMs can sometimes perform the task directly from instructions. The engineering effort is not eliminated, however. It reappears in new forms such as prompt design, demonstration selection, model choice, fine-tuning strategy, and evaluation design [4], [5].

C. Supervision and Adaptation

The reviewed papers also show a change in how supervision is used. Papers [1]-[3] rely primarily on labeled training data and task-specific model fitting. Their main question is which representation and classifier can learn the label mapping most accurately. Paper [1] adds an annotation question by using TextBlob to relabel or assist labeling. By contrast, paper [4] explicitly tests zero-shot and few-shot in-context learning, where the LLM may receive no in-domain training data. Paper [5] compares supervised fine-tuning of public models with in-context learning of proprietary GPT models. The central trade-off becomes labeled-data cost versus model scale and inference cost.

The findings suggest that the optimal supervision regime depends on data availability. When abundant in-domain labels are available, a smaller specialized model can be highly competitive. In paper [4], fully trained T5 remains clearly stronger than LLMs on structured ABSA. When only a few labeled examples are available, LLM in-context learning becomes more attractive because it can exploit prior knowledge from pretraining. Paper [4] reports large few-shot advantages over T5 trained on the same small number of examples. Paper [5] shows a related phenomenon across languages: English-centric public LLMs adapt quickly as target-language examples are added, while smaller multilingual models are particularly strong in zero-shot transfer.

D. Performance Trends and the Limits of Raw Score Comparison

Within-study evidence supports contextual pretraining, but not a single universal effect size. In [2], fine-tuned BERT improves accuracy by a modest 1–5 percentage points over the reported trigram and BiLSTM baselines. Paper [3] reports a much larger gap, but the baseline concerns discussed in Section IV.C make that result more useful as directional evidence than as a typical effect-size estimate. Taken together, the studies support the benefit of contextual pretraining while showing that the magnitude of the gain depends on experimental design.

The LLM studies further show that newer or larger models are not automatically better. In [4], ChatGPT is competitive on simpler sentiment classification but remains far behind fully trained T5-large on structured ABSA; Flan-UL2 performs well on SC and MAST but records 0.00 average on ABSA in the zero-shot table. In [5], multilingual smaller models can outperform much larger public English-centric LLMs when no target-language examples are available. When target-language examples are added through fine-tuning, public models improve substantially, whereas the same examples appended to GPT-4's prompt do not improve its average F1. These results point to task structure, language coverage, and adaptation method as key determinants of performance.

E. Computational Cost, Accessibility, and Deployment

Computational requirements increase substantially across the evolution. Lexicon and sparse-feature models can be trained or executed on ordinary hardware. Paper [2] explicitly contrasts BERT on Google Colab with fastText/BiLSTM and LSVM on a standard 16-GB Windows server, noting that the hybrid BiLSTM model is computationally cheaper while only one percentage point behind BERT in accuracy. This remains a relevant deployment consideration. A highly accurate model that requires cloud GPUs may be inappropriate for low-latency, private, offline, or resource-limited environments.

LLMs add another cost dimension because inference can require large GPU memory or paid API access. They also create reproducibility issues when proprietary models change over time. Paper [4] finds prompt sensitivity on complex tasks, while paper [5] uses different adaptation mechanisms for public models (supervised fine-tuning) and proprietary models (in-context prompting). Thus, practical comparison should include not only predictive performance but also training cost, inference cost, latency, privacy, model availability, and the ability to update the model for a changing domain.

F. Annotation Quality and Evaluation Design

Paper [1] demonstrates that annotation quality can dominate apparent performance. Its strongest scores are measured against automated TextBlob labels, so label provenance becomes part of the evaluation rather than a neutral preprocessing detail. A model can agree closely with a consistent labeling rule without establishing equivalent agreement with human judgment. This makes independent label validation essential when automated annotation is used.

Paper [4] identifies a parallel issue at the benchmark level. Different studies use different task definitions, prompts, output formats, and metrics, making broad statements about "LLM sentiment ability" difficult to justify. SENTIEVAL is proposed to create a more comprehensive and instruction-based evaluation. Paper [5] further shows that language and culture affect evaluation. A sentiment expression that is natural in English may not transfer directly to Spanish, French, or Chinese, and translated datasets may still differ from naturally produced target-language data. Evaluation design is part of the model-comparison problem, not a neutral afterthought.

G. Evidence-Quality Matrix

Table II applies five qualitative criteria to the reported experimental designs. The matrix is a review aid, not a formal risk-of-bias instrument; "Partial" indicates that a criterion is only partly satisfied or incompletely reported.

TABLE II. Qualitative evidence-quality matrix (✓ = strong, Partial = mixed/incomplete, ✗ = weak or not reported).

Criterion	[1]	[2]	[3]	[4]	[5]
Split / design	Partial	✓	✗	✓	✓
Baseline strength	✓	✓	✗	✓	✓
Label validity	✗	Partial	Partial	✓	Partial
Repeated runs / variance	✗	Partial	✗	Partial	✗
Comparable adaptation	✓	✓	Partial	Partial	Partial

Basis for ratings: [1] reports its highest scores against TextBlob-generated labels, and TextBlob agrees with human experts on only 491 of 1,000 separately annotated tweets. [2] states a 70/30 split and trains all four models on the same data. [3] does not state its split, reports single values without seeds or variance, and trains the Word2Vec-based classifiers with the same 10-epoch, 1×10^{-5} settings used for BERT fine-tuning. [4] uses established benchmark test sets and averages its smaller-model results over three seeds, but reports single zero-shot LLM runs; few-shot T5 is trained on the same sampled examples given to the LLMs, whereas the zero-shot comparison is against fully trained T5. [5] uses human-annotated English and Spanish data but GPT-4-translated French and Chinese data, reports single F1 values without seeds or variance, and compares fine-tuned public models with prompted GPT models.

The strongest cross-paper conclusions are methodological rather than numerical: label validity, task structure, adaptation strategy, and language coverage materially shape reported performance.

VI. DISCUSSION AND INSIGHTS

The central insight from the five papers is that the "LLM revolution" in sentiment analysis is a sequence of changes in representation, supervision, and interface rather than a sudden replacement of previous methods. Lexicon and n-gram systems rely on explicit rules or features; recurrent networks learn sequential representations; BERT adds pretrained contextual representations; and LLMs allow task behavior to be specified through instruction, fine-tuning, or in-context examples. Each stage removes some constraints while introducing new ones.

First, contextualization is a consistent improvement, but confidence should be weighted by experimental quality. Paper [2] provides the clearest estimate in this review because strong baselines remain close to BERT, whereas [3] supports the direction of the effect more strongly than its magnitude. The evidence-quality assessment in Table II makes this distinction explicit.

Second, LLMs change the relationship between model training and task use. A classical pipeline expects a fixed task, a labeled dataset, feature extraction, model training, and evaluation. A zero-shot LLM can accept a textual task definition and immediately classify new examples. This dramatically lowers the setup cost for a new task, especially when labeled data are scarce. Paper [4] demonstrates this advantage in few-shot settings, while [5] shows rapid adaptation across languages. However, ease of use should not be confused with universal superiority. Fine-tuned T5 remains stronger on complex structured tasks in [4], and smaller multilingual models provide excellent zero-shot cross-lingual transfer in [5].

Third, model scale is not a sufficient selection criterion. In [5], mT5-large (1.2B parameters) outperforms all four evaluated 7–8B public English-centric LLMs in zero-target-language transfer. In [4], Flan-UL2 shows strong zero-shot SC (77.31 average) and MAST (68.33) but scores 0.00 on structured ABSA, demonstrating that size and broad instruction tuning do not guarantee compliance with complex output structures. Pretraining distribution, task format, multilingual coverage, and adaptation strategy can matter as much as parameter count.

Fourth, the meaning of "better" changes across generations. In early classification studies, better often means higher accuracy or F1 on one dataset. In the LLM era, a system may be preferred because it can generalize across tasks, operate with few examples, support many languages, or avoid training a new model. Conversely, a smaller model may be preferred because it is cheaper, faster, easier to host privately, and more reproducible. The appropriate comparison depends on the intended use case.

Finally, the papers reveal that sentiment analysis remains an open problem despite high scores on standard benchmarks. Ambiguity, sarcasm, implied sentiment, structured aspect extraction, cultural variation, evolving language, domain shift, and label subjectivity remain difficult. Paper [4] shows that complex tasks still expose large gaps between LLMs and specialized models, and paper [5] shows uncertainty beyond the three tested target languages. The field has progressed from asking whether machines can classify positive and negative text to asking whether models can robustly understand subjective meaning across tasks, domains, languages, and changing social contexts.

A. A Practical Model-Selection View

Taken together, the five papers suggest a practical decision framework rather than a single winning algorithm. The relevant choice depends on five constraints: how stable the

task is, how much trustworthy labeled data are available, how much context must be modeled, which languages must be supported, and what deployment budget is acceptable. Table III translates the comparative findings into a model-selection guide. The recommendations are not universal rankings; they indicate which model family is most defensible under the resource conditions represented in the reviewed studies.

TABLE III. Practical model-selection guide derived from the reviewed evidence.

Deployment situation	**Likely starting point**	**Reason**
Stable narrow task; many labels	Compact ML / recurrent model	Low cost; easy retraining; strong task-specific fit
Context-sensitive task; moderate labels	BERT-style encoder	Contextual representation with controlled fine-tuning
Few labels; task changes often	Instruction-tuned LLM	Fast adaptation through prompting or limited examples
Low-resource multilingual transfer	mT5 / XLM-R, then fine-tune if possible	Multilingual pretraining can beat larger English-centric LLMs
Strict privacy, latency, or offline use	Locally hosted compact model	Predictable cost, reproducibility, and data control

For a stable, high-volume sentiment task with a mature label set, a compact supervised system remains a rational first choice. Paper [2] is especially instructive: the hybrid fastText-BiLSTM model reaches 90% accuracy, only one point below fine-tuned BERT, while running on a conventional server rather than the cloud environment used for BERT. In a production pipeline that processes millions of similar messages, that small predictive difference may be less important than predictable latency, lower infrastructure cost, easier retraining, and local data control. Classical TF-IDF/SVM models can also remain valuable as transparent baselines and fallback systems when interpretability or hardware constraints dominate.

When sentiment depends strongly on context, BERT-style encoders provide a useful middle ground. They offer pretrained bidirectional representations without requiring the scale or open-ended generation behavior of an LLM. The review supports this choice most clearly when comparison baselines are strong: [2] shows a modest but consistent BERT improvement over trigram and BiLSTM alternatives. A BERT encoder is therefore attractive when organizations have enough labeled examples for fine-tuning, need stable class outputs, and can afford more compute than a sparse or recurrent model but do not need a general-purpose generative interface.

Instruction-tuned LLMs become more attractive when the label set changes frequently, annotation is scarce, or rapid prototyping is more important than minimizing inference cost. Paper [4] shows that LLMs can use instructions and a small number of examples to perform sentiment tasks without building a separate classifier from scratch. The benefit is strongest when the alternative would be training a specialized model from very little data. The limitation is that prompt design and output format become part of the system, and complex structured tasks can still favor a smaller model that is directly fine-tuned for the required output.

Multilingual deployment requires a separate decision rather than simply selecting the largest model. In [5], mT5-large is the strongest public model when no target-language examples are available. Once target-language labels are introduced, several fine-tuned public models improve sharply, whereas adding the same examples to GPT-4 as in-context demonstrations provides little benefit. This suggests a practical sequence for low-resource languages: begin with a multilingual pretrained model, collect a small amount of reliable target-language data, and fine-tune if the deployment permits parameter updates.

Finally, hybrid designs remain relevant across generations. Paper [1] combines lexicon-based annotation with supervised models, paper [2] combines fastText with BiLSTM, and paper [3] combines BERT representations with neural classifiers. A modern system can likewise separate components for labeling assistance, representation, classification, and human review instead of forcing one model to perform every function. The broader implication is that model selection should optimize a system-level objective—validated sentiment quality under a defined cost, language, privacy, and update constraint—rather than maximize a benchmark score in isolation.

VII. LIMITATIONS OF THE REVIEW

This review is focused rather than exhaustive. Five recent studies were selected to represent distinct stages of sentiment-analysis development, so the conclusions identify patterns in this evidence rather than constituting a systematic census of the field. The studies also differ in domain, label scheme, task complexity, metric, and adaptation regime, which limits direct numerical synthesis. For that reason, the report emphasizes within-study comparisons and the qualitative evidence assessment in Table II.

A second limitation is that experimental reporting is uneven across the five papers. Some studies specify splits, baselines, and evaluation settings clearly, while others omit information needed to assess sensitivity to initialization, data partitioning, or optimization. Proprietary models and proprietary call-center data further restrict exact replication. These differences constrain the certainty of cross-study conclusions even when the reported numbers are internally correct.

Finally, language coverage remains narrow. The broad multi-task study [4] is English-only, and [5] evaluates three target languages under a design that mixes naturally sampled and translated data. The evidence is therefore informative about the tested settings but should not be generalized to multilingual sentiment analysis as a whole.

VIII. RESEARCH GAPS AND FUTURE DIRECTIONS

One research gap is human-grounded and uncertainty-aware evaluation. Future sentiment benchmarks should document annotation procedures, inter-annotator agreement, class balance, and the handling of ambiguous cases. Automated labeling can reduce cost, but its outputs should be independently validated and clearly distinguished from human ground truth.

A second gap is robust evaluation across domains and naturally occurring languages. Future work should include more naturally authored low-resource language data, code-

switching, dialects, slang, and culturally specific indirect expression. When machine translation is used to create evaluation data, the translation model should be separated from the evaluated system where feasible, or the dependency should be disclosed and analyzed.

A third gap is fair comparison across adaptation regimes. Studies should define “zero-shot” and “few-shot” precisely, report the number and language of all examples available to each model, and compare methods under matched annotation and compute budgets where possible. Prompt templates, random seeds, model versions, inference settings, and repeated-run statistics should also be reported so that conclusions remain reproducible under paraphrased instructions and alternative data splits.

A fourth gap concerns efficiency and deployment. Future studies should report training time, inference latency, memory use, compute or energy budget, and cost per prediction alongside accuracy or F1. This is especially important when a compact fine-tuned model is within a few percentage points of a much larger system. Privacy, version stability, API availability, and the ability to host a model locally should also be treated as model-selection criteria.

A fifth gap is continual adaptation to evolving language. Sentiment expressions change with new products, communities, events, slang, and social conventions. Zhang et al. [4] describe a major limitation of current LLMs as their lack of flexibility in fine-tuning or re-training, while [5] shows that a small amount of target-language supervision can materially improve adaptable public models. Future systems should combine broad pretrained knowledge with efficient domain and language updates while preserving reliability on previously learned tasks.

IX. CONCLUSION

This review examined five recent studies to trace the evolution of sentiment analysis from lexicon-based and traditional feature pipelines to recurrent deep learning, BERT-based Transformers, and modern LLMs. The evidence shows a clear progression in how text is represented and how models are adapted. TextBlob and sparse features provide simple and efficient baselines; LSTM, GRU, and BiLSTM systems learn sequential representations; BERT adds contextual bidirectional representations; and LLMs enable instruction-based zero-shot and few-shot use across a wider range of tasks.

The transition is meaningful but not absolute. BERT shows a consistent advantage over local or static representations in the reviewed conventional studies, but the most defensible effect size is modest when the comparison baseline is strong. In the LLM studies, performance depends strongly on task structure, language, labeled-data availability, and adaptation method. Specialized or multilingual smaller models can outperform larger LLMs, and [5] shows that target-language examples help much more when used for parameter-updating fine-tuning than when appended to GPT-4’s prompt. Architecture size alone is an inadequate model-selection criterion.

The most important broader lesson is that modern sentiment analysis is no longer only a classification-accuracy problem. Reliable systems require attention to annotation quality, contextual validity, computational cost, prompt stability, multilingual coverage, privacy, and evaluation design. The LLM era has made sentiment analysis more flexible and accessible, especially when labeled data are scarce, but it has also created new sources of variability. Future progress is likely to depend on combining the strengths of specialized efficient models with the adaptability and broad pretrained knowledge of larger language models.

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